

ParaSail: A Retrieval-Augmented Geospatial Decision Support System for Sustainable Fishing, Marine Conservation and Coastal Livelihood Resilience

26th Digital Blue Economy Summit | Research Level: Experimental Paper

Anantha Krishnan AS · Naipunnya School of Management, Cherthala, India

September 2026

Abstract

Small-scale and artisanal fishing communities in data-poor coastal regions plan their trips on fragmented information: weather applications carry no fisheries context, habitat research is rarely operational, vessel-tracking analytics are locally inaccessible, and conservation rules remain static documents. This paper presents ParaSail(Predictive Advisory and Retrieval-Augmented System Advancing Informed Livelihoods), a production-style, retrieval-augmented geospatial decision support system that fuses live meteorological and oceanographic forecasts, biodiversity occurrence records, satellite imagery and regulatory texts into a single explainable advisory. The system follows a five-layer architecture (ingestion, processing, prediction, retrieval-augmented explanation, advisory), enforces marine protected areas and seasonal closures as hard constraints, and scores every recommendation through a transparent sustainability index combining catch likelihood, weather safety and bycatch risk. A retrieval layer over a dual-collection vector database attaches regulation, guidance and traditional-knowledge context to every advisory, while parameter-efficient LoRA fine-tuning (rank 8; under one percent trainable parameters) enables local adaptation on modest hardware. A case study on the southwest coast of India, covering monsoon closures, a protected-area network and active sardine and mackerel fisheries, demonstrates exact closure enforcement, monotonic scoring behaviour and topically relevant retrieval. The work shows that sustainability-scored, explainable fishing advisories are feasible today with open-source tools, and that conservation, legality and safety can be embedded as constraints in the objective function rather than bolted on afterwards.

Keywords: decision support system; retrieval-augmented generation; geospatial artificial intelligence; LoRA fine-tuning; fisheries sustainability; marine conservation; blue economy; explainable AI

1. Introduction

1.1 Background and Crisis Context

The blue economy, understood as the sustainable use of ocean resources for economic growth, improved livelihoods and employment while preserving the health of marine ecosystems, has become a central policy frame for coastal states [1]. Marine capture fisheries sit at its heart: small-scale fisheries alone supply at least 40 percent of the global catch and support the livelihoods, nutrition and culture of nearly half a billion people [2]. Yet the resource base is eroding. The Food and Agriculture Organization reports that 35.5 percent of assessed stocks were fished beyond biologically sustainable levels in 2019, the highest share ever recorded [3]. Pollution compounds the pressure: roughly 11 million tonnes of plastic enter the ocean every year and are projected to nearly triple by 2040 without decisive action [4], [5].

Climate change is redistributing the resource itself. Projections indicate large-scale shifts in maximum catch potential as species track moving thermal habitats [6], and on the southwest coast of India the oil sardine, historically the backbone of the regional pelagic fishery, has shown pronounced fluctuations and range extension associated with oceanographic variability [7]. Traditional ecological calendars calibrated to historical seasons lose predictive value when spawning times, aggregation zones and migration routes move. The past is no longer a reliable map of the sea.

Management reform could recover substantial value: rebuilding overfished stocks under sound governance is projected to lift global catches well above business-as-usual trajectories [8], while biodiversity loss demonstrably undermines the ecosystem services that fisheries depend upon [9], [10]. The open question is not whether sustainable management is worthwhile, but whether coastal actors possess the information infrastructure to act on it in real time.

1.2 Problem Statement

No integrated, real-time and explainable decision support system exists for safe, legal and sustainable fishing in data-poor coastal regions. The consequences are fourfold. Economically, fuel, time and labour are wasted on blind search for shifting fish distributions. Ecologically, fishing pressure concentrates on spawning grounds, sensitive habitats and protected zones. In terms of human safety, small vessels put to sea without granular marine wind and wave guidance. And institutionally, the information that does exist is fragmented across tools that never speak to one another: weather applications carry no fisheries context, habitat studies remain academic rather than operational, AIS analytics are priced and structured for global commodity fleets rather than local ones, and regulations live in static documents that are difficult to query.

1.3 Research Gap and Questions

The research gap is architectural rather than algorithmic. Individual capabilities exist, from numerical weather prediction and satellite ocean colour to species distribution models and vessel detection, but no operational system fuses weather, ocean state, species distributions, closures and ecological risk into a single advisory with explainable reasoning. This paper therefore addresses four questions. RQ1: Can open environmental APIs, biodiversity records and satellite imagery be fused into one normalised, quality-controlled geospatial data model? RQ2: Can conservation rules, including protected areas and seasonal closures, be enforced as hard constraints on every recommendation? RQ3: Can every advisory carry retrievable, human-readable justification drawn from regulations, guidance and local knowledge? RQ4: Can the resulting system be adapted to new regions and species without code changes and within modest hardware budgets?

1.4 Contributions

The contributions of this work are as follows.

- C1** A five-layer reference architecture (ingestion, processing, prediction, retrieval-augmented explanation, advisory) for fisheries decision support, implemented end to end with open-source components.

- C2** A rules engine that enforces marine protected areas and seasonal closures as hard, geographically exact constraints ahead of any scoring.
- C3** A transparent sustainability index $S = 0.45C + 0.25W + 0.30(1 - B)$ with four advisory classes, monotonic in every input by construction.
- C4** A dual-collection retrieval-augmented layer, spanning text (regulations, closures, guidance, local knowledge) and satellite tiles, that attaches verifiable context to every advisory.
- C5** A LoRA-based fine-tuning pathway (rank 8 on query and value projections; under one percent trainable parameters) enabling region and species adaptation on mid-range GPUs.
- C6** An operational validation across live-fetch, rule-enforcement, scoring-behaviour and retrieval-relevance test families on a southwest-India case study.

The remainder of the paper reviews related work (Section 2), states the objectives (Section 3), details the architecture and technology (Section 4) and the methodology (Section 5), describes the data and case study (Section 6), reports results (Section 7), discusses applications and impact (Section 8), limitations (Section 9), future scope (Section 10) and conclusions (Section 11), and closes with the summit's five mandatory questions (Section 12).

2. Related Work

2.1 Operational Fisheries Advisory Services

The most mature precedent is India's Potential Fishing Zone (PFZ) advisory service, which derives satellite sea-surface-temperature and chlorophyll fronts to suggest likely aggregation zones to fishers [11]. Such services demonstrate both demand and viable delivery channels, but they answer only the question of where fish may be: they carry no legality constraints, no bycatch or ecological-risk dimension, no weather-safety gate, and no explanation attached to individual recommendations. ParaSail treats PFZ-style frontal signals as one input among several inside a constrained, explainable advisory.

2.2 Species Distribution and Habitat Modelling

Correlative species distribution models relate occurrence records to environmental predictors and remain the workhorse for estimating habitat suitability, including in data-poor settings [12]. They estimate where a species could be, not whether fishing there is legal, safe or ecologically responsible, and they are rarely wired into operational decision flows. This system accordingly uses a random-forest habitat model as the fallback estimator for data-poor taxa, precisely because it is inexpensive, interpretable and robust to modest occurrence counts, while learned spatiotemporal models handle data-rich species.

2.3 Deep Learning for Marine Monitoring

Deep learning has transformed marine image analysis: single-stage detectors in the YOLO family support real-time catch and bycatch recognition [13], Vision Transformers set accuracy benchmarks for fine-grained species classification [14], and recurrent architectures such as the LSTM model spatiotemporal sequences [15]. Classical ensembles remain competitive baselines for tabular

environmental data [16]. ParaSail deploys one representative of each family, selected for the constraints of coastal deployment rather than leaderboard position, as detailed in Section 4.4.

2.4 Retrieval-Augmented Generation and Parameter-Efficient Fine-Tuning

Retrieval-augmented generation grounds model outputs in retrieved documents, measurably improving factual verifiability in knowledge-intensive tasks [17], and has become the standard pattern for question answering over evolving corpora [18]. The underlying transformer machinery [19] is expensive to fine-tune wholesale; LoRA freezes pretrained weights and injects low-rank updates, reducing trainable parameters by orders of magnitude at matched task performance [20]. Sentence encoders in the Sentence-BERT family provide efficient embeddings for retrieval over regulatory text [21], and contrastive vision-language models such as CLIP enable embedding-based search over unlabelled satellite imagery [22]. The present work combines these techniques into a dual-collection retrieval layer and a parameter-efficient adaptation pathway.

2.5 Traditional Ecological Knowledge and Digital Humanities

Traditional ecological knowledge, contextual, cumulative and locally grounded, has long been recognised as a legitimate knowledge system for fisheries governance [23]. Digital humanities methods for recording oral histories and calendars give such knowledge a durable, retrievable digital form. Here the retrieval corpus treats local knowledge as a first-class source, so that an advisory can cite both statutory rules and community experience.

2.6 Positioning

Table 1 positions the system against the tool classes fishers actually encounter. The differentiator is not any single capability but the fusion: constraints applied before scores, explanations attached to outputs, and adaptation achieved without re-engineering.

Table 1 Positioning against existing tool classes

Tool class	Strengths	What is missing
Weather applications	Granular wind and wave forecasts	No fish, closures or bycatch dimension
PFZ-style advisories [11]	Satellite-derived location hints	No legality, safety gate or explanations
Habitat / SDM studies [12]	Rigorous suitability estimates	Not operational, not integrated
AIS fleet analytics	Fleet-scale effort insight	Costly, legally restricted, not local
Static regulations	Authoritative rules	Unqueryable, no geospatial linkage
ParaSail (this work)	Fused, constrained, explainable	Longitudinal field trial (future work)

3. Research Objectives

The primary objective is to design, implement and evaluate a retrieval-augmented geospatial decision support system that recommends safe, legal and ecologically responsible fishing windows. This decomposes into eight objectives, each of which maps onto a validation test family (Section 5.2).

RO1 Ingest and normalise live forecasts, wave and wind fields, sea-surface temperature, chlorophyll-a and current fields from open APIs and satellite archives.

- RO2** Model aggregation likelihood and movement tendency for registered species, with habitat-suitability fallback for data-poor taxa.
- RO3** Enforce marine protected areas, seasonal closures and bycatch proxies as hard or soft constraints on every recommendation.
- RO4** Implement a retrieval layer storing regulations, conservation guidance and local ecological knowledge, attached as context to every advisory.
- RO5** Support local adaptation through custom data ingestion and parameter-efficient LoRA fine-tuning.
- RO6** Expose the system through a production-style API and a visual dashboard.
- RO7** Evaluate through live-fetch, rule-enforcement, scoring-behaviour and retrieval-relevance tests.
- RO8** Align the design with blue-economy principles, treating livelihoods, conservation and safety as joint objectives.

4. System Architecture and Technology

4.1 Architectural Overview

ParaSail is organised as five layers: ingestion, processing, prediction, retrieval-augmented explanation and advisory, each independently replaceable and configured through YAML. Data flows sequentially: open APIs, satellite archives and community uploads are validated and normalised onto a common spatiotemporal grid; predictive models estimate catch likelihood and movement tendency; the retrieval layer assembles regulation and context; and the advisory layer applies constraints before scoring, emitting a machine- and human-readable advisory with retrieved citations. PostgreSQL with PostGIS holds the relational and geospatial state, Qdrant holds vector embeddings, and Docker Compose orchestrates deployment.

4.2 Ingestion Layer

Each provider is wrapped behind a common interface with per-source freshness contracts, response caching and an offline contingency: when a live endpoint is unavailable, the system degrades to the most recent cached fields, flags their age in the advisory and keeps operating. Table 2 summarises the principal sources.

Table 2 Principal data sources and their roles

Source	Content	Cadence	Role in advisory
Open-Meteo [24]	Hourly meteorological and marine forecasts (wind, waves)	Hourly	Weather-safety component W
Copernicus Marine [25]	Near-real-time ocean analysis (SST, currents)	Daily	Environmental features
NASA ERDDAP	SST and chlorophyll-a fields	Daily	Environmental features
GBIF / OBIS [26], [27]	Historical species occurrences	Static corpus	Model training, habitat baselines
FishBase / WoRMS [28]	Taxonomy, traits, seasonality	Static corpus	Species registry, name normalisation

Source	Content	Cadence	Role in advisory
Sentinel-2 / Sentinel-3 [29], [30]	Multispectral satellite tiles	Per pass	Satellite retrieval collection

4.3 Processing Layer

The processing layer enforces coordinate and timestamp validation, normalises taxonomy against WoRMS identifiers, regrid heterogeneous products onto a common spatiotemporal grid, and engineers features such as thermal anomalies, chlorophyll fronts and wind stress. Quality gates reject malformed or stale inputs, and a freshness audit records the age of every field consumed by an advisory so that degraded data is always visible to the user rather than silently absorbed.

4.4 Prediction Layer and Model Selection

The prediction layer deliberately mixes model families, matching each to data availability and deployment constraints. Table 3 records the selected models and the rationale for each choice. For data-rich pelagic species, an LSTM over 24 hourly steps estimates short-horizon movement tendency; for data-poor taxa, a random-forest habitat-suitability model over environmental features provides a robust fallback; a YOLOv8 detector supports catch-image verification; and a Vision Transformer classifier, adapted with LoRA, recognises species in field imagery. LoRA freezes the pretrained weights and learns a low-rank additive update,

$$W = W_0 + \Delta W, \Delta W = BA, B \in \mathbb{R}^{d \times r}, A \in \mathbb{R}^{r \times k} \quad (1)$$

where the rank $r = 8$ updates are injected into the query and value projections of all twelve encoder blocks. For ViT-B/16 this leaves under one percent of parameters trainable, approximately 0.29 million of 85.8 million, which keeps adaptation within mid-range GPU memory.

Table 3 Selected models and selection rationale

Task	Selected model	Rationale
Catch-image detection	YOLOv8n, Ultralytics [13]	Real-time on modest hardware; mature toolchain
Species classification	ViT-B/16 + LoRA ($r = 8$) [14], [20]	Fine-grained accuracy; under 1% trainable parameters
Movement tendency (24 h)	Two-layer LSTM, hidden size 128 [15]	Short-horizon sequences for data-rich species
Habitat suitability	Random forest [16]	Interpretable fallback for data-poor taxa
Text embeddings	all-MiniLM-L6-v2 [21]	Fast 384-dimensional sentence embeddings
Satellite tile embeddings	OpenCLIP ViT-B/32 [22]	Similarity search over unlabelled tiles
Advisory explanation (optional)	Local instruct LLM (e.g. Qwen2.5-7B)	Summarises retrieved context; optional

4.5 Retrieval-Augmented Explanation Layer

Explainability is treated as retrieval rather than as generation of unconstrained prose. The Qdrant vector database [31] maintains two collections. The text collection holds chunked regulations, closure calendars, conservation guidance and recorded local ecological knowledge, embedded with a sentence encoder. The satellite collection holds OpenCLIP embeddings of Sentinel tiles annotated with bounding boxes and acquisition metadata. At advisory time the engine performs hybrid retrieval,

combining bounding-box and temporal pre-filtering with top-k vector search, and attaches the retrieved passages and tiles to the advisory payload, so that every recommendation carries its own justification. Beyond explainability, this design is what makes the corpus scalable for future data integration: new regulations, guidance documents, satellite passes and entire regional corpora are added by chunking, embedding and upserting, with no model retraining. Frequently changing knowledge, such as closure calendars, lives in a retrievable corpus rather than frozen in model weights, so updates take effect as data rather than code, and the vector store scales horizontally as sources grow.

4.6 Advisory Layer

The advisory layer first applies hard constraints: a request inside an active marine protected area or seasonal closure is refused outright with the blocking reason, before any scoring occurs. Remaining requests are scored on the sustainability index

$$S = 0.45C + 0.25W + 0.30(1 - B) \quad (2)$$

where C is the catch-likelihood composite, W the weather-safety component and B the bycatch and ecological-risk proxy. C blends the learned and habitat estimates through a data-availability weight,

$$C = \lambda \hat{C} + (1 - \lambda) \hat{H} \quad (3)$$

where $\hat{C}$ is the model-predicted aggregation index, $\hat{H}$ the habitat-suitability estimate, and λ rises with the volume and recency of occurrence data for the species. The weights in Eq. (2) reflect a deliberate normative choice: catch opportunity contributes less than the sum of the safety and ecological terms. They are exposed as configuration rather than hidden constants, so that governance bodies, not developers, own the trade-off. Table 4 defines the four advisory classes.

Table 4 Advisory classes and decision thresholds

Class	Condition	Interpretation
PROCEED	$S \geq 0.75$	Favourable conditions and low risk
PROCEED WITH CAUTION	$0.60 \leq S < 0.75$	Acceptable with vigilance
DELAY OR RELOCATE	$0.40 \leq S < 0.60$	Degraded weather or elevated bycatch risk
DO NOT FISH	$S < 0.40$, or any hard constraint	Unsafe, illegal or ecologically irresponsible

4.7 Deployment and Reproducibility

The stack comprises FastAPI for the API, PostgreSQL with PostGIS for relational and geospatial storage, Qdrant for vectors and Docker Compose for orchestration, fronted by a Tailwind, Leaflet and Chart.js dashboard. Deployment is configuration-driven: region, species registry, closure calendars, model weights and score weights are all specified in YAML, which is what makes region and species retargeting possible without code edits. Version control and containerisation make every reported result reproducible from a clean environment.

5. Methodology

5.1 Research Design

The study follows an experimental-paper design: iterative engineering organised into six phases, each closed by an explicit acceptance gate that must pass before the next phase begins (Table 5). This phase-gate discipline is what converts a prototype into an evaluable system: every capability claim is tied to a gate, and every gate to a test.

Table 5 Development phases and acceptance gates

Phase	Focus	Acceptance gate
P1	Governance, region selection, regulatory inventory	Region and rules inventory signed off
P2	Ingestion pipeline, database schema	Live fetches validated; freshness audit passes
P3	Baseline predictive models	Baselines beat naive persistence
P4	LoRA fine-tuning, custom data ingestion	Training converges within GPU memory budget
P5	RAG integration, advisory API	Closure and MPA blocks exact; top-k retrieval relevant
P6	Dashboard, pilot feedback loop	Advisory schema legible to pilot users

5.2 Evaluation Protocol

Four test families operationalise the research questions. T1 (live fetch) exercises end-to-end ingestion across four coastal cities, asserting valid payloads and freshness within contract. T2 (rule enforcement) requires that requests inside the June-July sardine closure and inside a reserve are blocked with the correct reason. T3 (scoring behaviour) requires that synthetic input sweeps show the index monotonic in every argument, with class transitions exactly at the configured thresholds. T4 (retrieval relevance) requires that top-k retrieved context for closure, protected-area, weather-safety and bycatch queries is topically relevant and correctly attributed.

5.3 Ethics, Licensing and Data Governance

All ingested sources are open-licensed or publicly documented. Local-knowledge ingestion is consent-gated and attributed at source. The optional AIS module is disabled by default because vessel-effort data is legally restricted in many jurisdictions. The system is positioned as advisory: it recommends, never commands, and it displays data age so that degraded advice is identifiable as such.

6. Data Sources and Case Study

6.1 Case Study Region

The case study targets the southwest coast of India, selected for three properties: an active seasonal closure regime, including the June-July monsoon fishing closure and spawning protection for the oil sardine; a network of marine protected areas; and economically dominant small pelagic fisheries, sardine and mackerel above all, whose landings fluctuate strongly from year to year [7]. Four coastal cities anchor the live-fetch tests, spanning the length of the study coast.

6.2 Species Registry

Table 6 lists the registry. *Sardinella longiceps* (Indian oil sardine) is the pelagic mainstay and subject to the June-July closure; *Rastrelliger kanagurta* (Indian mackerel) is the co-dominant pelagic;

Thunnus tonggol (longtail tuna) represents a higher-value, wider-ranging target; and *Penaeus monodon* (giant tiger prawn) represents a data-poor, high-value taxon that exercises the habitat fallback pathway.

Table 6 Species registry for the case study

Species	Common name	Fishery role	Modelling path
<i>Sardinella longiceps</i>	Indian oil sardine	Pelagic mainstay; June-July closure	LSTM tendency + habitat blend
<i>Rastrelliger kanagurta</i>	Indian mackerel	Co-dominant pelagic	LSTM tendency + habitat blend
<i>Thunnus tonggol</i>	Longtail tuna	Higher-value pelagic	Habitat-led with occurrence priors
<i>Penaeus monodon</i>	Giant tiger prawn	Data-poor, high-value	Habitat-suitability fallback

6.3 Regulatory Corpus and Optional AIS Module

The regulatory corpus comprises protected-area polygons loaded into PostGIS, closure calendars expressed as species-month rules, and textual guidance chunked for retrieval. The optional AIS vessel-effort reference module remains disabled pending legal review, as noted in Section 5.3.

7. Results and Findings

7.1 Live Ingestion and Graceful Degradation

Across the four-city live-fetch tests, the ingestion layer returned valid, complete payloads with per-source freshness within contract. Under simulated outage the offline contingency engaged: advisories continued to be issued from cached fields with explicit age flags, demonstrating graceful degradation rather than failure. Operating costs remained negligible because every source is free at the volumes used.

7.2 Rule Enforcement

Rule enforcement was exact. A request for the oil sardine during the June-July window returned allowed = false, cited spawning protection as the reason and issued the DO NOT FISH class; coordinates inside a reserve were blocked regardless of otherwise favourable scores. Because constraints execute ahead of scoring, no configuration of the score weights can legalise an illegal trip, a property the tests confirmed under weight perturbation.

7.3 Scoring Behaviour and Monotonicity

Sweeps over each input confirmed monotonicity: S increases linearly in C and W and decreases linearly in B, with class transitions exactly at 0.40, 0.60 and 0.75. Moderate conditions keep the weather component above the small-vessel safety threshold, while degraded wind or wave inputs push the advisory to DELAY OR RELOCATE; high catch likelihood with low bycatch risk yields PROCEED or PROCEED WITH CAUTION. The decision surface is legible to stakeholders: a fisher can see which term moved the class.

7.4 Retrieval Relevance and Explainability

For closure, protected-area, weather-safety and bycatch queries, the top-k retrieved context was topically relevant, and the dashboard attaches it to each advisory, satisfying the explainability requirement. Retrieval quality was verified by inspection against the source corpus rather than by automatic ranking metrics alone, an appropriate standard at the current stage of system maturity.

7.5 Engineering Findings

LoRA fine-tuning of the classifier converged within mid-range GPU memory with under one percent of parameters trainable, confirming the adaptation pathway. Region and species retargeting was exercised entirely through configuration, without code changes. The advisory schema proved simultaneously machine-parseable, for the dashboard, and human-readable, for pilot feedback. Table 7 summarises the validation outcomes.

Table 7 Validation outcome summary

Test family	Checks performed	Outcome
T1 live fetch	Four coastal cities; payload validity; freshness	Reliable; graceful offline degradation
T2 rule enforcement	June sardine closure; in-reserve coordinates	Exact blocks with correct reasons
T3 scoring behaviour	Monotonicity sweeps; threshold transitions	Monotonic in every input; classes as configured
T4 retrieval relevance	Closure, MPA, weather and bycatch queries	Topically relevant context attached

8. Applications and Impact

For fishers, the system promises reduced search time and fuel spend, safer trips and fewer accidental violations. For fisheries departments, it offers an evidence base for advisory bulletins and environmental monitoring. For conservation organisations, it operationalises compliance as a built-in constraint rather than an enforcement afterthought. For researchers, the ingestion and modelling pipelines are reusable infrastructure for distribution and climate-impact studies. For coastal communities, the retrieval corpus offers a durable digital home for traditional ecological knowledge, oral histories and fishing calendars, a concrete digital-humanities contribution [23].

The work aligns primarily with Sustainable Development Goal 14 (Life Below Water), particularly its targets on sustainable management and effective regulation of harvesting, and secondarily with Goals 2, 8, 9 and 13; Table 8 maps the mechanisms.

Table 8 Alignment with the Sustainable Development Goals

Goal	Mechanism in this system
SDG 14 Life Below Water	Closures and MPAs as hard constraints; bycatch term in the index
SDG 2 Zero Hunger	Improved catch-per-effort for nutritious small pelagics
SDG 8 Decent Work	Reduced fuel and search labour; safer working conditions
SDG 9 Industry and Innovation	Open-source, low-cost advisory infrastructure
SDG 13 Climate Action	Climate-adaptive, data-driven fishing calendars

The normative contribution deserves emphasis. Conservation, legality and safety are embedded in the objective function itself: the system cannot recommend an illegal or protected-area trip under any weight configuration, and ecological risk enters the score on equal footing with catch opportunity. Responsible marine AI, on this evidence, is less a matter of model scale than of where the constraints sit in the pipeline.

9. Limitations

Predictive skill is bounded by data quality: occurrence records carry sampling bias, and satellite fields inherit sensor and cloud-cover uncertainty. Data-poor taxa rely on habitat envelopes and expert rules rather than learned dynamics. Forecast horizons beyond a few days compound uncertainty, so advisories are probabilistic guidance, never guarantees. AIS- and VMS-derived effort data remains legally restricted in many jurisdictions and is therefore optional. Most importantly, no longitudinal field trial has yet been conducted: adoption, trust and realised catch and safety outcomes remain unmeasured.

10. Future Scope

Five directions follow naturally. Longitudinal pilots with fishing cooperatives, measured against pre-system baselines, would test real-world impact. Coupling to ocean-circulation and climate-projection models would extend horizons for anticipating distribution shifts. Federated learning across regions would adapt models without exchanging sensitive data. Blockchain traceability linking advisories to catch records would support verified sustainable seafood. And multilingual, voice-first, offline-capable mobile clients would extend reach to non-literate users and connectivity-poor vessels.

11. Conclusion

This paper delivered ParaSail, a production-grade, retrieval-augmented geospatial decision support system for the digital blue economy. It integrated live environmental data, biodiversity records, satellite imagery and conservation rules into one coherent architecture, and its operational validation showed that sustainability-scored, explainable advisories are feasible with open-source tools and modest hardware. The core lesson is that responsible marine AI embeds conservation, legality and safety as constraints, and earns trust through explainability and local adaptation. The work thereby satisfies the summit's key principle: digital technology addressing a real problem of fisheries, oceans, coastal communities, conservation and sustainability.

12. The Five Mandatory Summit Questions

- Q1** What is the problem? Fishers and coastal managers lack a real-time, integrated, explainable advisory system for sustainable fishing; the consequences are wasted effort, accidental violations, ecological damage and avoidable safety risk.
- Q2** Why is it important? It affects livelihoods, food security, marine conservation and human safety simultaneously, which is precisely the intersection the blue economy must govern.
- Q3** What digital technology is used? Geospatial AI (YOLOv8, Vision Transformer with LoRA, LSTM, random forest), retrieval-augmented generation over a dual-collection vector database, and open live environmental APIs, orchestrated with FastAPI, PostGIS, Qdrant and Docker.
- Q4** What data and evidence supports the study? Open meteorological, oceanographic, biodiversity and satellite data, regulation datasets, and a working system validated through live-fetch, rule-enforcement, scoring-behaviour and retrieval-relevance tests on a southwest-India case study.

Q5 What solution and contribution is provided? A production-grade platform issuing sustainability-scored, explainable fishing advisories while enforcing conservation rules as hard constraints, adaptable to new regions and species through configuration and parameter-efficient fine-tuning.

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